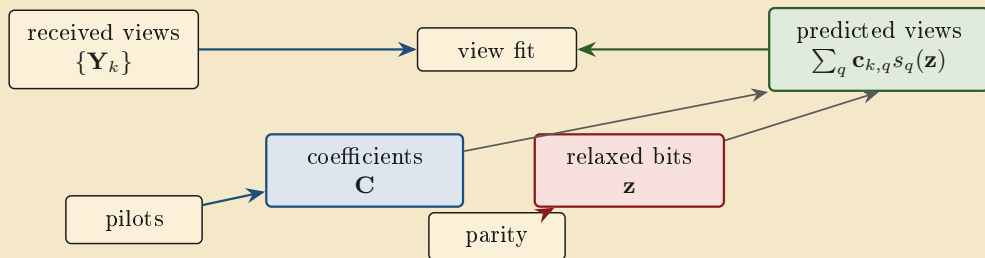


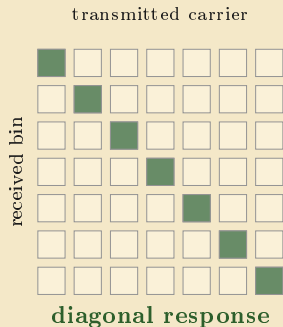
The Joint Demodulation and Decoding Problem

Carrier 4 inside the frame-wide receiver



The combiner is a read-out. The joint problem scores the coefficients and relaxed bits against the received views.

One-tap model



$$\mathbf{Y}^{\text{full}} = \mathbf{H}_{\text{fd}} \mathbf{X} + \boldsymbol{\nu} \quad (2)$$

adequate cyclic prefix

$$L_{\text{cp}} \geq L_h - 1$$

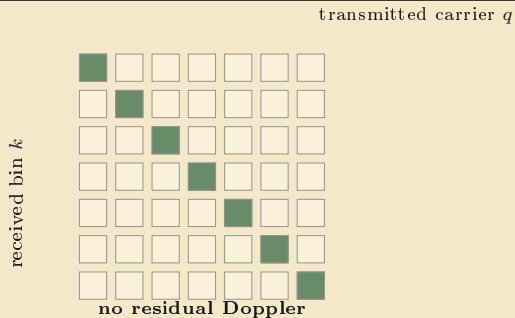
channel does not vary
over the useful interval

one channel gain acts on each carrier:

$$Y_k^{\text{full}} = H_{k,k} X_k + \nu_k$$

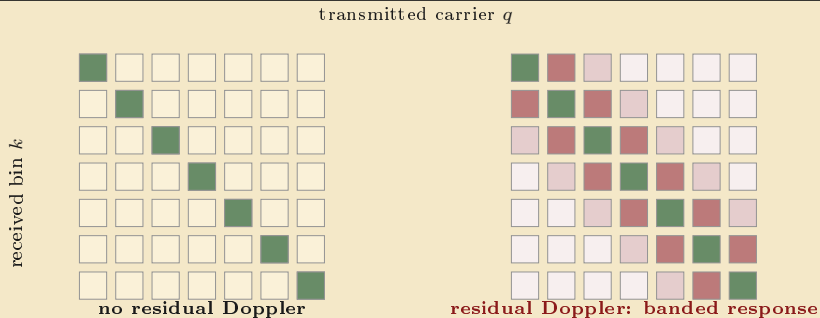
Under both conditions, the fast Fourier transform diagonalizes the block channel and one pilot-specified equalizer tap per carrier suffices.

Same-symbol ICI



The same symbol index with a different carrier index gives intercarrier interference (ICI). Residual Doppler makes the frequency-domain operator non-diagonal. For small residual Doppler and carrier-frequency offset, the coefficients remain near the diagonal and motivate a banded operator.

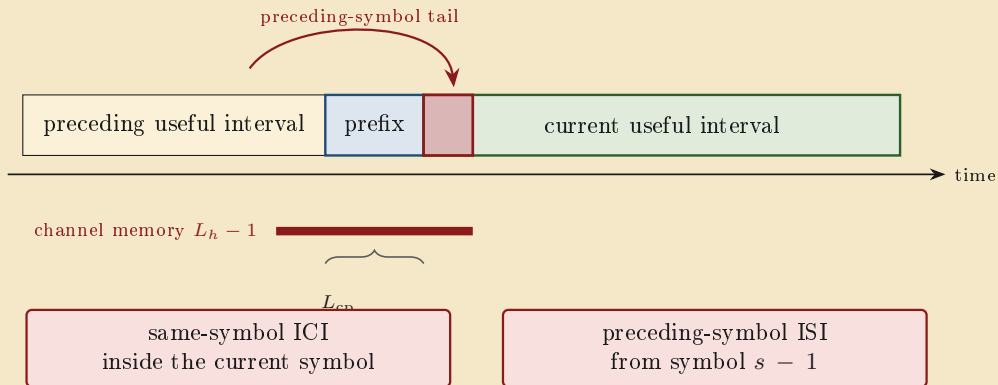
Same-symbol ICI



$$Y_k^{\text{full}} = H_{k,k} X_k + \sum_{q \in \mathcal{K} \setminus \{k\}} H_{k,q} X_q + \nu_k \quad (5)$$

The same symbol index with a different carrier index gives intercarrier interference (ICI). Residual Doppler makes the frequency-domain operator non-diagonal. For small residual Doppler and carrier-frequency offset, the coefficients remain near the diagonal and motivate a banded operator.

Insufficient cyclic prefix

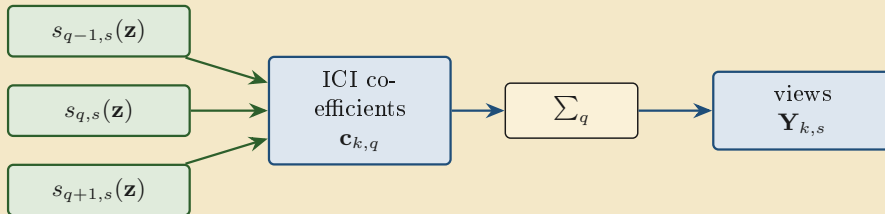


A tap beyond the prefix carries the preceding symbol into the useful interval. Different carriers in one symbol give ICI; different symbol indices give ISI.

Model scope

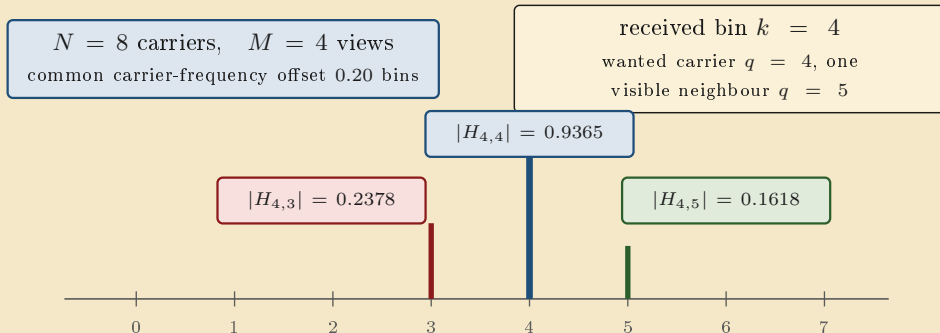
preceding symbol $\mathbf{X}_{s-1}$

$$\mathbf{Y}_{k,s} = \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_{q,s}(\mathbf{z}) + \nu_{k,s} \quad (28)$$



The current coefficient model represents the same-symbol response. The preceding-symbol term requires the adjacent-symbol extension in the backup. The reported results do not validate that extension.

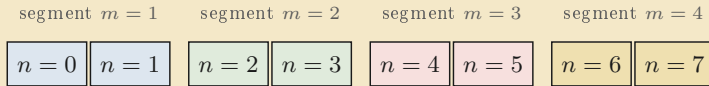
Worked example



With zero offset, this unit-gain toy gives $\mathbf{H} = \mathbf{I}$.

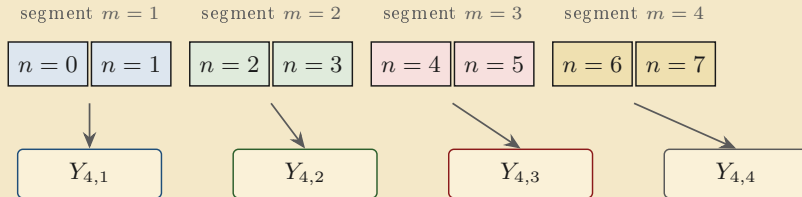
The running example keeps carrier 4, its four views, and carrier 5 visible; the complete calculation retains all eight carriers. It uses one OFDM symbol for an ICI example, not a measured Red channel or an ISI example.

Partial-FFT views



Partial-FFT creates no new information. It keeps temporally localized components of the full-interval statistic that one full FFT would collapse.

Partial-FFT views

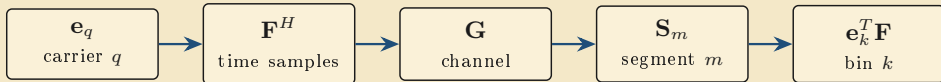


$$Y_{k,m} = \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{r}, \quad \mathbf{Y}_k = [Y_{k,1}, \dots, Y_{k,M}]^T \in \mathbb{C}^M \quad (7)$$

For this example, $\mathbf{Y}_4 \in \mathbb{C}^4$ contains four windowed-FFT values from the same eight-sample OFDM block.

Partial-FFT creates no new information. It keeps temporally localized components of the full-interval statistic that one full FFT would collapse.

ICI coefficients



$$c_{k,q,m} \triangleq \mathbf{e}_k^T \mathbf{F} \mathbf{S}_m \mathbf{G} \mathbf{F}^H \mathbf{e}_q \quad (25)$$

one coefficient for received bin k ,
transmitted carrier q , and view m

$$Y_{k,m} = \sum_{q \in \mathcal{K}} c_{k,q,m} X_q + \nu_{k,m} \quad (27)$$

every view is a complete carrier mix

$$\mathbf{Y}_k = \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) + \boldsymbol{\nu}_k, \quad \mathbf{c}_{k,q} = [c_{k,q,1}, \dots, c_{k,q,M}]^T \quad (28)$$

The receiver estimates the ICI coefficients; it does not evaluate the time-domain operator $\mathbf{G}$. The band retains nearby carrier pairs, while the complete toy calculation keeps all eight carriers.

Coefficient sum

m	$c_{4,4,m}$	$c_{4,5,m}$
1	$0.2485 + 0.0196j$	$0.1985 + 0.1011j$
2	$0.2303 + 0.0954j$	$-0.1575 + 0.1575j$
3	$0.1895 + 0.1619j$	$-0.1011 - 0.1985j$
4	$0.1302 + 0.2125j$	$0.2200 - 0.0348j$
sum	$0.7985 + 0.4893j$	$0.1598 + 0.0253j$

$$\sum_{m=1}^M c_{k,q,m} = H_{k,q} \quad (29)$$

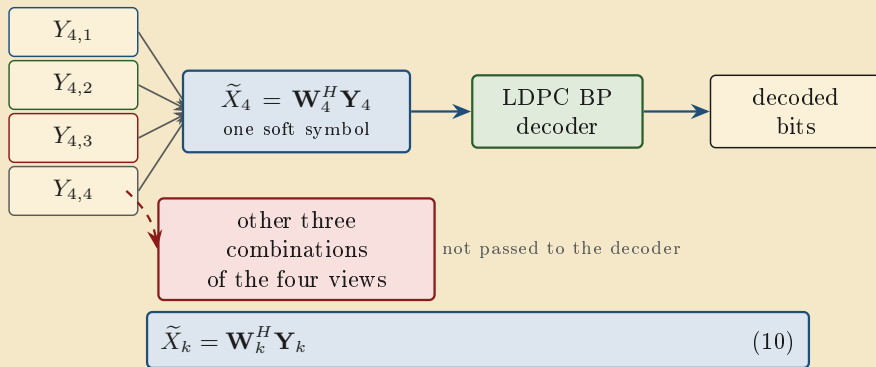
$$\text{when } \sum_{m=1}^M \mathbf{S}_m = \mathbf{I}$$

maximum difference over all pairs
 2.2×10^{-16}

windowed FFT: $Y_{4,1} = -0.1396 + 0.3878j$ complete
 eight-carrier mix: $\sum_{q=0}^7 c_{4,q,1} X_q = -0.1396 + 0.3878j$
 difference 1.1×10^{-16}

The per-view coefficients and the channel coefficient are the same channel at two resolutions. The visible $q = 4$ and $q = 5$ columns do not assert that the other carrier pairs are zero.

The read-out

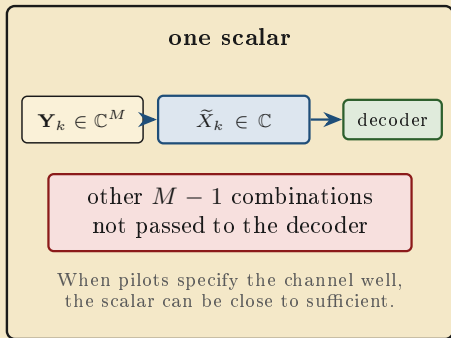


The read-out keeps one combination of a carrier's views for the decoder. The ICI coefficients are $\mathbf{C}$; the post-reception combiner is $\mathbf{W}$.

Scalar and vector

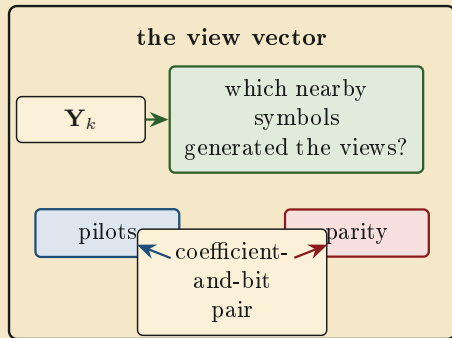
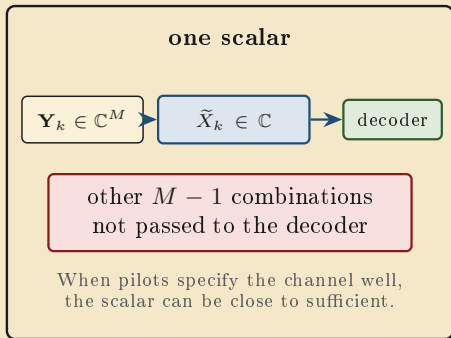
Is one scalar enough?

Scalar and vector



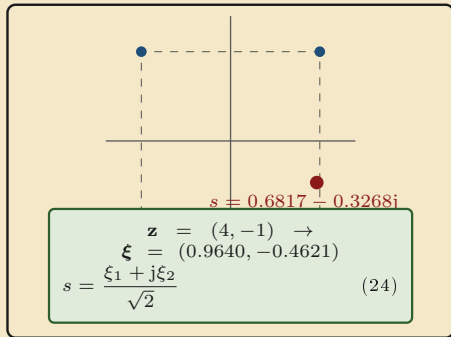
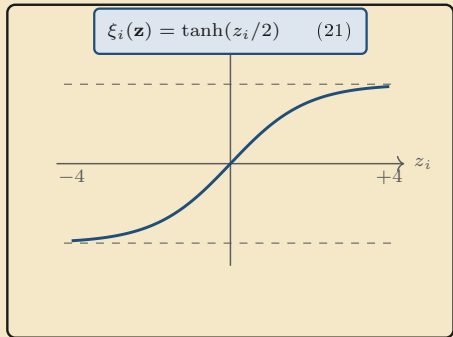
The scalar is sufficient exactly when $p(\mathbf{b} \mid \mathbf{Y}) = p(\mathbf{b} \mid \tilde{\mathbf{X}})$. The joint search instead keeps the view vectors; pilots constrain the coefficients and parity checks restrict the coded bits.

Scalar and vector



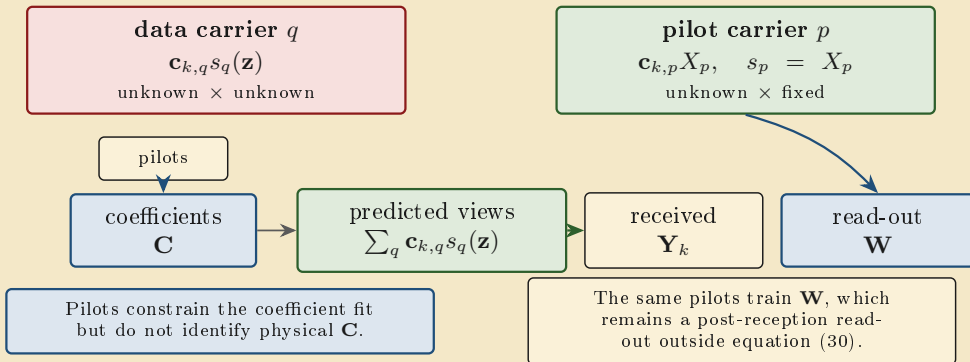
The scalar is sufficient exactly when $p(\mathbf{b} \mid \mathbf{Y}) = p(\mathbf{b} \mid \tilde{\mathbf{X}})$. The joint search instead keeps the view vectors; pilots constrain the coefficients and parity checks restrict the coded bits.

Relaxed QPSK



Hard bits give the four quadrature phase-shift keying points. Relaxed bits move the symbol inside the square and provide a differentiable path. Pilot carriers keep their known symbols.

Pilot constraints



Parity penalty

$$\mathcal{R}_{\text{par}}(\mathbf{z}) = \sum_j \left(1 - \prod_{i \in \mathcal{N}_j} \xi_i(\mathbf{z}) \right)^2 \quad (22)$$

where $\mathcal{N}_j$ is the set of bits in parity check j .

hard bits

even parity

product +1.0000

penalty 0.0000

relaxed values

correct signs

product +0.4032

penalty 0.3561

one sign changed

wrong parity

product -0.4032

penalty 1.9691

The penalty vanishes on codewords. Relaxed values with the correct signs remain costly until their magnitudes approach one.

Joint objective

received views
signal fit

Joint objective

received views
signal fit

coefficients $\mathbf{C}$
coefficient energy

Joint objective

received views
signal fit

coefficients $\mathbf{C}$
coefficient energy

relaxed bits $\mathbf{z}$
parity penalty

Joint objective

received views
signal fit

coefficients $\mathbf{C}$
coefficient energy

relaxed bits $\mathbf{z}$
parity penalty

$$\mathcal{J}(\mathbf{C}, \mathbf{z}) = \underbrace{\lambda_d \sum_{k \in \mathcal{K}} \left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) \right\|_2^2}_{\text{signal term}} + \underbrace{\gamma_C \|\mathbf{C}\|_F^2}_{\text{coefficient prior}} + \underbrace{\lambda_p \mathcal{R}_{\text{par}}(\mathbf{z})}_{\text{parity penalty}} \quad (30)$$

λ_d , γ_C , and λ_p weigh the view fit, coefficient energy, and parity penalty.
The view noise and coefficient prior are isotropic complex Gaussian.

Pilots constrain $\mathbf{C}$ through the signal term. Parity checks constrain $\mathbf{z}$. The combiner $\mathbf{W}$ remains outside the objective.

Separate checks

signal check

true coefficient-and-bit pair

noise $\sigma = 0.05$

0.0185

one parity check

hard even: 0

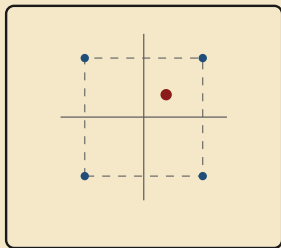
relaxed signs: 0.3561

one wrong sign: 1.9691

These are separate checks. They do not form a complete numerical $\mathcal{J}$ and do not share a scale.

The signal term rewards a pair that reproduces the views. The parity term rewards relaxed bits that harden toward a codeword.

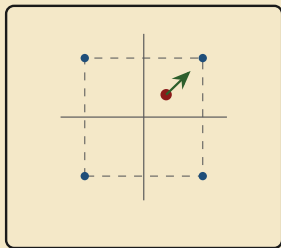
Local gradient check



$$t = 0: z_4 = (0.8000, 0.8000) \\ s_4 = 0.2687 + 0.2687j$$

Local check of mean-only equation (30): $\mathbf{C}$ is held at its true values; $\mathbf{z}$ starts with the correct signs.
One plain fixed-step update moves only $\mathbf{z}$; it is not an implemented receiver update.
This one step is not a general descent guarantee.

Local gradient check

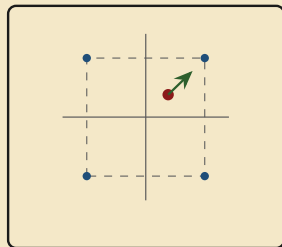


$t = 0: z_4 = (0.8000, 0.8000)$
 $s_4 = 0.2687 + 0.2687j$

gradient at z_4
 $(-0.2638, -0.2451)$
step size $\eta = 0.8$
move $(+0.2110, +0.1961)$

Local check of mean-only equation (30): $\mathbf{C}$ is held
at its true values; $\mathbf{z}$ starts with the correct signs.
One plain fixed-step update moves only $\mathbf{z}$; it is not an implemented receiver update.
This one step is not a general descent guarantee.

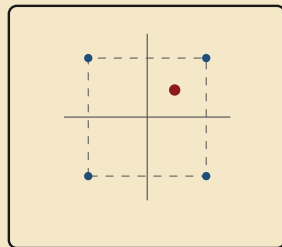
Local gradient check



$t = 0: z_4 = (0.8000, 0.8000)$
 $s_4 = 0.2687 + 0.2687j$

gradient at z_4
 $(-0.2638, -0.2451)$
step size $\eta = 0.8$
move $(+0.2110, +0.1961)$

$\mathcal{J} : 3.5608 \rightarrow 2.6663$
 $z_4 : (0.8000, 0.8000) \rightarrow$
 $(1.0110, 0.9961)$



$t = 1: z_4 = (1.0110, 0.9961)$
 $s_4 = 0.3298 + 0.3257j$

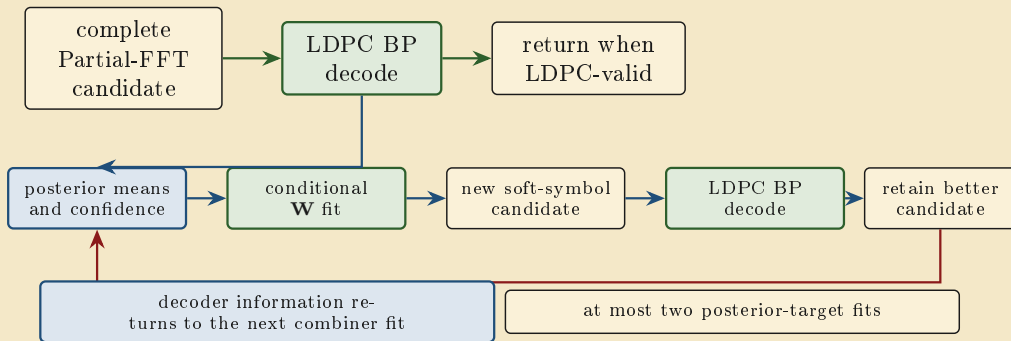
Local check of mean-only equation (30): $\mathbf{C}$ is held
at its true values; $\mathbf{z}$ starts with the correct signs.
One plain fixed-step update moves only $\mathbf{z}$; it is not an implemented receiver update.
This one step is not a general descent guarantee.

Five receivers

OFDM+LDPC sum views	outer-pilot equalization $\rightarrow$ LDPC BP	payload
Partial-FFT+LDPC keep views	pilot-trained $\mathbf{W} \rightarrow$ one soft symbol $\rightarrow$ LDPC BP	payload
JUNA-Iterative keep views	decoder means return $\rightarrow$ conditional $\mathbf{W}$ fit	payload
JUNA-(C, z) keep views	solve $\mathbf{C} \mid \mathbf{z} \rightarrow$ move $\mathbf{z}$	initialization only
JUNA-Direct-(C, z) keep views	move $\mathbf{C}, \mathbf{z}$ together $\rightarrow$ derive $\mathbf{W}$	initialization only

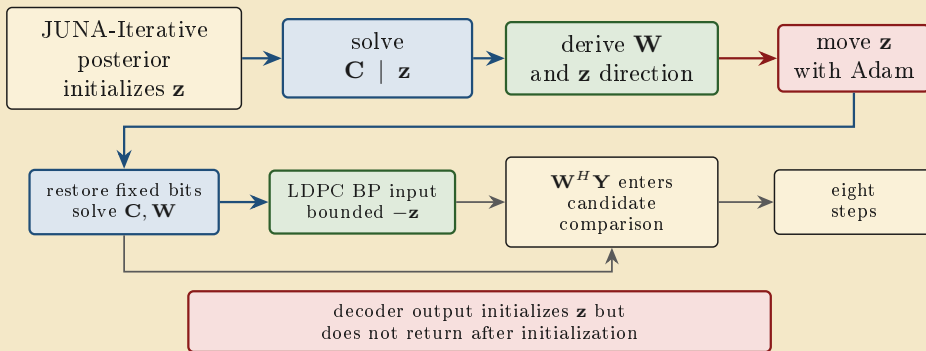
The first two receivers separate demodulation from decoding. The next three use decoder information or relaxed coded bits during symbol estimation.

JUNA-Iterative



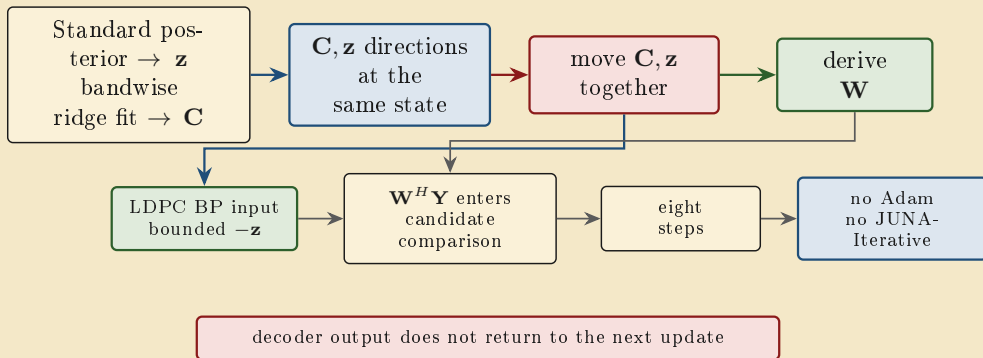
Like turbo equalization, JUNA-Iterative repeats equalization and decoding. This exchange approximates the joint loop, but forms no $\mathbf{C}$, has no free $\mathbf{z}$, and never evaluates equation (30).

JUNA- (C, z)



The coefficient and relaxed-bit blocks are motivated by equation (30). The executed frame cost is given with the experiment settings and is not identical to equation (30).

JUNA-Direct- (C, z)



JUNA-Direct- (C, z) moves the coefficient and relaxed-bit variables together. The weights are derived from the updated coefficients and remain outside equation (30).

Takeaway

residual Doppler
→ same-symbol ICI

Partial-FFT
keeps the received views

one scalar
can discard combinations

pilots constrain
 $\mathbf{C}$

parity checks constrain
 $\mathbf{z}$

equation (30) scores
the pair in the views

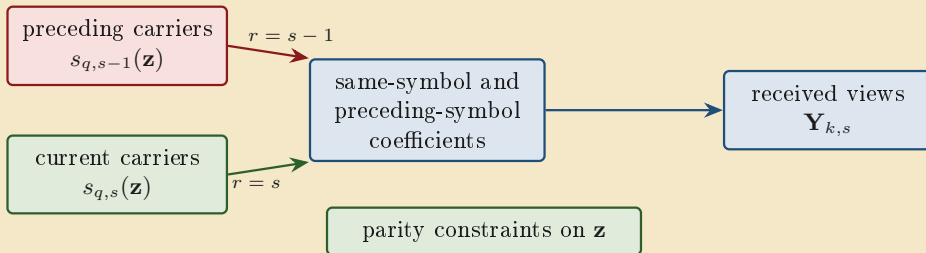
Only JUNA-Iterative returns de-
coder information to another update.

$\mathbf{W}$ is the post-reception read-out
and remains outside equation (30).

The current objective represents the same-symbol response. An insufficient cyclic prefix also creates a preceding-symbol term; representing that term requires the adjacent-symbol extension in the backup.

Adjacent-symbol extension

$$\mathbf{Y}_{k,s} = \sum_{(q,r) \in \mathcal{B}(k,s)} \mathbf{c}_{k,s;q,r} s_{q,r}(\mathbf{z}) + \mathbf{n}_{k,s} \quad (\text{B1})$$



Here $\mathcal{B}(k, s)$ contains the represented carrier and symbol indices. This is a modeling extension, not a short-cyclic-prefix experiment. The reported results do not validate this extension.

Posterior-variance alternative

$$\begin{aligned} X_q &= \mu_q + \varepsilon_q \\ \mu_q &= s_q(\mathbf{z}), \quad \mathbb{E}[\varepsilon_q] = 0 \\ \mathbb{E}[\varepsilon_p^* \varepsilon_q] &= 0, \quad p \neq q \end{aligned}$$

$$\begin{aligned} v_q &= \mathbb{E}|\varepsilon_q|^2 \\ \text{QPSK: } v_q &= 1 - |\mu_q|^2 \end{aligned}$$

$$\bar{\mathcal{J}}_{\text{sig}} = \lambda_d \sum_{k \in \mathcal{K}} \left[\left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} \mu_q \right\|_2^2 + \sum_{q \in \mathcal{B}_K(k)} v_q \|\mathbf{c}_{k,q}\|_2^2 \right]$$

The first term fits the symbol means. The second retains their scalar variance. Pilots and fixed carriers use $v_q = 0$.

This alternative is not used for the reported mean-only JUNA- (C, z) results. It is not a marginalized likelihood and omits parity-induced correlations.

The posterior-variance signal term replaces only the signal term of equation (30).